

Dipak Kumar Biswal

Engineering Lead | AI Application Engineering

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PROFESSIONAL SUMMARY

Frontend Engineering Lead with 8.5+ years specialising in React 18/19, TypeScript, and high-performance web architecture. Currently expanding into **AI Application Engineering** — with hands-on experience integrating LLM features, agentic AI summarisation, and Vercel AI SDK into production healthcare applications. Led WinForms → React 18 modernisation earning written client commendation; delivered MRF-as-a-Service (Machine-Readable Files) as sole developer; reduced production defects by 40% through TDD and mentoring. Targeting roles at the intersection of frontend excellence and AI-native product development.

CORE COMPETENCIES

Frontend & Architecture	React 18/19 · TypeScript · Redux Toolkit · Jotai · React Query · TanStack · Vite · Modular Design · Performance Optimisation
UI & Styling	Chakra UI · Material UI · Kendo UI · Styled-Components · SASS/LESS
Quality & DevOps	TDD (Jest + RTL) · 92% Test Coverage · GitHub Actions · Docker · Agile/Scrum · Code Review
AI & Emerging Tech	LLM Integration · Agentic AI · LangChain/LangGraph · Vercel AI SDK · Multi-Agent Systems · Prompt Engineering
RAG & Vector Search	Retrieval-Augmented Generation (RAG) · Vector Databases (Pinecone / Chroma / pgvector) · Embeddings · Semantic Search · Document Chunking & Indexing
Languages	TypeScript (expert) · Python (actively learning for agentic AI) · JavaScript
Domains	Healthcare (Health Edge, US Fertility) · Finance · Manufacturing · Logistics

PROFESSIONAL EXPERIENCE

Persistent Systems Ltd

Hyderabad

Engineering Lead | July 2022 – Present

WEB UI Modernisation — Health Edge

- Led migration of legacy WinForms Manager application to React 18 + TypeScript; reduced load time from 8s → 1.8s (77% improvement) earning written client commendation.
- Integrated an **AI Summariser** feature using React 18 + LLM integration — auto-generating actionable summaries of complex healthcare documents, negotiated rates, and invoices; directly accelerated user decision-making.
- Established TDD culture with Jest + React Testing Library; achieved **92% test coverage** and cut production defects by **40%**.
- Mentored junior engineers and conducted 30+ code reviews, promoting modular, testable, and maintainable code practices.

MRF-as-a-Service — Health Edge (Sole Developer)

- Designed and delivered full-stack Machine-Readable Files (MRF) web app from scratch: React + TypeScript + Chakra UI + Jotai + React Query — enabling dynamic rate generation, invoice repricing, and scheduled background processes.
- Optimised with Vite, lazy loading, and TanStack caching → achieved **1.2s average response time**.
- Implemented comprehensive Jest + RTL test suite (92% coverage); internally recognised and recommended for core UI modernisation team.

US Fertility — Client Project

- Developed responsive patient-navigation web app using React.js, Material UI, and Redux Toolkit.
- Collaborated with cross-functional teams to integrate UI/UX improvements and optimise data flows for healthcare outcomes.

Hinduja Tech Limited

Bangalore

Senior Consultant | Oct 2019 – July 2022

- Delivered multiple high-impact SPAs including Direct Store Delivery (DSD) and SAP UI5-based Leave Management System (LMS) for education clients.

- Built contract management dashboard for seed-production industry; implemented safety workflows for ADNOC Workforce Management System (WMS).
- Owned full delivery lifecycle: requirement gathering, feasibility analysis, stakeholder communication, and customer support.

L&T; Technology Services

Bangalore

Senior Engineer | Sep 2018 – Sep 2019

- Built SPAs for manufacturing change orders and tool exchange management using HTML5, Bootstrap, jQuery.
- Implemented secure authentication (LDAP + JWT), SMTP notifications, and D3.js data visualisations.
- Authored design/requirement documents; created Jasmine + Karma test suites for full coverage.

Global Edge Software Ltd

Bangalore

Software Engineer | May 2017 – Sep 2018

- Provided application support and maintenance for Android-based mobile applications, resolving bugs and handling minor enhancements across multiple client engagements.
- Assisted in UI development and troubleshooting for manufacturing-focused Android apps, contributing to stable production releases.
- Gained cross-client exposure across 3 projects within a year, building adaptability in fast-switching delivery environments.

EDUCATION

Bachelor of Technology (B.Tech) — Mechanical Engineering

May 2014

BPUT University, Odisha

CERTIFICATIONS & CONTINUOUS LEARNING

- **IBM RAG and Agentic AI Professional Certificate** — Coursera / IBM | *In Progress (3/10 courses completed)* — covering RAG pipelines, multi-agent systems, LangChain, LangGraph, CrewAI, and Model Context Protocol (MCP)
- Self-directed study: Python for Agentic AI · Async/Concurrency · LLM APIs · Tool Calling · Vector Databases · Prompt Engineering

ADDITIONAL

Languages:English (professional) · Hindi · Odia (native)

Interests:Reading · Teaching · AI/ML research · Rubik's Cube